

Four-Point Exponential Metric Kernels and Minimal Five-State Obstructions to Markov Embedding

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Abstract

Let (X, d) be a finite metric space and let $K(d)$ be its exponential metric matrix,

$$K(d)_{ij} = e^{-d(i,j)}.$$

For a positive matrix K , Perron normalisation produces a stochastic matrix diagonally similar to $K/\lambda_{\max}(K)$. We study when this stochastic snapshot is the unit-time member of a homogeneous continuous-time Markov semigroup. The problem is equivalent, in the favourable cases considered here, to asking whether an ordinary matrix logarithm of K is Metzler; this is distinct from the classical entrywise/Hadamard notion of infinitely divisible positive-definite kernels.

We prove that for every metric on at most four points the principal logarithm of $K(d)$ has nonnegative off-diagonal entries. Consequently every four-point exponential metric kernel is strongly infinitely divisible in the ordinary matrix-root sense, and its Perron-normalised stochastic matrix is Markov embeddable. The proof is direct: every four-point metric is written as a rectangle with nonnegative pendant lengths, adjacent resolvent signs are controlled exactly, opposite entries are reduced by pendant monotonicity, and a reciprocal involution closes the remaining integral sign.

At five points the property fails. We give a three-parameter deleted-corner family obtained from a 3×2 product metric. Its matrix logarithm has a strictly negative retained cross-corner entry. We then prove that the kernel has simple positive spectrum for all interior parameters, including the only degenerate Walsh case, so the Perron-normalised stochastic matrix has a unique real logarithm. The negative logarithm entry therefore excludes every homogeneous Markov embedding, not merely the principal or reversible construction. Hence five is the exact minimum cardinality at which an exponential metric kernel can exhibit this obstruction. A divisor-overlap kernel yields an infinite arithmetic subfamily, including the five-state set $\{1, 2, 3, 4, 6\}$. Finally, under common attenuation $K_s = (1 - s)I + sK$, we prove that the obstructing logarithm entry has exactly one sign crossing. We stop at this boundary: a complete classification of five-state exponential metric kernels remains open.

Keywords: Markov embedding; matrix logarithm; exponential metric kernel; strongly infinitely divisible matrix; Metzler matrix; stochastic matrix; matrix roots; finite metric space.

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1 Introduction

The Markov embedding problem asks whether a stochastic matrix T can be represented as

$$T = e^G$$

for a continuous-time Markov generator G , that is, a real matrix with zero row sums and nonnegative off-diagonal entries. A foundational treatment was given by Kingman [1]; it is closely tied to stochastic matrix roots, generator identifiability, and one-parameter semigroups. Modern treatments include matrix-root methods [5], the connection with nonnegative infinite divisibility and M -matrices [7], algebraic analyses of the embedding problem [8, 10], and recent work on arbitrarily fine stochastic divisibility [11].

This paper studies a structured source of stochastic matrices. Given a finite metric $d = (d_{ij})$, define the entrywise exponential metric matrix

$$K(d)_{ij} = e^{-d_{ij}}, \quad d_{ii} = 0. \quad (1)$$

When $K(d)$ is positive definite, it has a well-defined real symmetric principal logarithm. If the off-diagonal entries of $\log K(d)$ are nonnegative, then $K(d)$ is the exponential of a Metzler matrix. Perron normalisation then converts the same logarithm, up to diagonal similarity and a scalar diagonal shift, into a Markov generator. This gives a direct bridge between metric geometry, ordinary matrix roots, and Markov embeddability.

There is a terminology collision worth fixing at the outset. In the classical theory of positive-definite kernels, an “infinitely divisible kernel” often means that all positive *Hadamard* powers remain positive semidefinite; this is the setting of Schoenberg and Horn [3, 4]. Here, following the nonnegative-matrix and Markov-embedding literature [7, 13], *strong infinite divisibility* refers to ordinary matrix multiplication: an invertible nonnegative matrix A has a nonnegative ordinary m th root for every $m \in \mathbb{N}$. These are different operations. Our core assertions concern the ordinary matrix exponential e^B , the ordinary matrix logarithm $\log A$, and ordinary matrix roots.

The general embedding problem is already delicate in low dimension. Casanellas, Fernández-Sánchez and Roca-Lacostena showed that the principal-logarithm test is not valid for arbitrary Markov matrices: there are embeddable 4×4 matrices whose principal logarithm is not a Markov generator [9]. They subsequently gave a detailed treatment of the general 4×4 embedding problem [10]; a later systematic account is given by Baake and Sumner [15]. Thus a negative off-diagonal entry in a principal logarithm is not, by itself, a certificate of non-embeddability. When the stochastic matrix has pairwise distinct positive real eigenvalues, however, the real logarithm is unique, and the principal-logarithm test is decisive; this is a classical consequence of real logarithm theory [2] and is also explicit in modern embedding analyses [9]. Reversible embedding has its own sharper structure [6, 12].

The results below isolate a sharp threshold inside the structured family (1):

1. Every exponential metric kernel on at most four points has a Metzler principal logarithm. After Perron normalisation it is homogeneously Markov embeddable.
2. There is a three-parameter family of five-point exponential metric kernels for which the principal logarithm has a strictly negative off-diagonal entry and the spectrum is positive and simple. Hence no alternative real logarithm can repair the sign, and the Perron-normalised stochastic matrix is not embeddable.

3. Therefore five is the exact minimum state cardinality at which this phenomenon occurs.
4. In the deleted-corner family, common attenuation produces exactly one sign crossing of the obstructing logarithm entry.

The first result is not a replacement for the general 4×4 embedding theory. Rather, it identifies a geometrically defined subfamily for which embeddability is automatic and supplies a direct metric proof. The second result shows that this automatic property fails immediately at the next cardinality, even inside a simple product-metric construction. The resulting $4/5$ threshold is the main structural statement of the paper.

The arithmetic motivation comes from the divisor-overlap kernel

$$q(n, m) = \frac{\tau(\gcd(n, m))}{\sqrt{\tau(n)\tau(m)}}. \quad (2)$$

Writing $x_p(n) = \log(\nu_p(n) + 1)$ gives

$$q(n, m) = \exp\left(-\frac{1}{2} \sum_p |x_p(n) - x_p(m)|\right), \quad (3)$$

so every finite divisor-overlap restriction is an exponential ℓ^1 -metric kernel. An earlier preprint developed this kernel in a broader arithmetic setting and recorded the five-state restriction $\{1, 2, 3, 4, 6\}$ using its principal logarithm [14]. The present paper isolates the embedding problem, corrects the logical scope by proving real-logarithm uniqueness for the obstruction, generalises the example to a parameter family, and proves the exact minimal cardinality.

2 Preliminaries: Perron normalisation, logarithms, and divisibility

A real matrix B is *Metzler* if $B_{ij} \geq 0$ for $i \neq j$. A matrix Q is a *Z-matrix* if $Q_{ij} \leq 0$ for $i \neq j$, so B is Metzler exactly when $-B$ is a Z-matrix.

Definition 2.1 (Markov embeddability). *A row-stochastic matrix T is homogeneously Markov embeddable if there exists a real matrix G such that*

$$G\mathbf{1} = 0, \quad G_{ij} \geq 0 \ (i \neq j), \quad T = e^G.$$

Such a G is a continuous-time Markov generator.

Definition 2.2 (Strong infinite divisibility). *Following Van-Brunt [7], an invertible nonnegative matrix A is strongly infinitely divisible if for every $m \in \mathbb{N}$ there is a nonnegative matrix A_m with $A_m^m = A$.*

Van-Brunt proved that a nonnegative matrix A is strongly infinitely divisible if and only if $A = e^B$ for some Metzler matrix B (equivalently $A = e^{-Q}$ for a Z-matrix Q) [7]. For the positive result we use the elementary direction: if $A = e^B$ with B Metzler, then $A_m = e^{B/m} \geq 0$. The converse is recalled only as background; a direct odd-root argument proves corollary 4.4. For the general characterisation, the author subsequently supplied a corrected proof [16].

Lemma 2.3 (Perron normalisation). *Let K be a strictly positive real symmetric matrix, let $\lambda > 0$ be its Perron eigenvalue, and let $r > 0$ be a corresponding Perron vector. Put $R = \text{diag}(r)$ and*

$$T = R^{-1}(K/\lambda)R. \quad (4)$$

Then T is row-stochastic and reversible with stationary weights proportional to r_i^2 . If K is positive definite, then

$$\log T = R^{-1}(\log K - \log(\lambda)I)R. \quad (5)$$

Consequently, if $\log K$ is Metzler, then $\log T$ is a Markov generator and T is embeddable.

Proof. Since $Kr = \lambda r$,

$$T\mathbf{1} = R^{-1}(K/\lambda)r = \mathbf{1}.$$

Moreover

$$r_i^2 T_{ij} = \lambda^{-1} r_i r_j K_{ij} = r_j^2 T_{ji},$$

so detailed balance holds. Equation (5) follows from analytic functional calculus under similarity and from the scalar factor λ^{-1} . Positive diagonal similarity preserves signs of off-diagonal entries. Finally $\log T \mathbf{1} = 0$ because $T\mathbf{1} = \mathbf{1}$ and the eigenvalue 1 lies on the principal branch. Thus a Metzler $\log K$ yields a Markov generator. $\square$

The negative direction requires more care because the principal logarithm need not be the only real logarithm.

Lemma 2.4 (Simple positive spectrum closes the branch loophole). *Let K be real symmetric positive definite with pairwise distinct eigenvalues, and let T be its Perron normalisation (4). Then T has pairwise distinct positive real eigenvalues and a unique real logarithm, namely its principal logarithm. Therefore, if any off-diagonal entry of $\log K$ is negative, T is not Markov embeddable.*

Proof. Diagonal similarity and multiplication by λ^{-1} preserve simplicity and positivity of the spectrum. Let G be a real matrix satisfying $e^G = T$. Then G commutes with T because every matrix commutes with its exponential. Since T has simple real spectrum, each one-dimensional real eigenspace of T is invariant under G , so in the real eigenbasis of T the matrix G is diagonal with real entries g_i . The equations $e^{g_i} = \lambda_i(T) > 0$ force $g_i = \log \lambda_i(T)$. Hence $G = \log T$. This is the simple-positive-spectrum case of the classical uniqueness theory for real matrix logarithms [2]. If $[\log K]_{ij} < 0$, then (5) gives $[\log T]_{ij} < 0$, so the unique real logarithm cannot be a Markov generator. $\square$

For a symmetric positive-definite matrix A we use the standard integral identity

$$\log A = \int_0^\infty \left(\frac{1}{1+t} I - (A + tI)^{-1} \right) dt. \quad (6)$$

This identity follows by applying the scalar identity to an orthogonal spectral decomposition of A . The matrix integral converges: the integrand is bounded near zero and is $O(t^{-2})$ at infinity. Thus for $i \neq j$,

$$[\log A]_{ij} = - \int_0^\infty (A + tI)_{ij}^{-1} dt. \quad (7)$$

3 Every four-point exponential metric kernel is embeddable

We begin with a self-contained representation of an arbitrary four-point metric.

Lemma 3.1 (Rectangle-plus-pendants decomposition). *Let d be a metric on $\{1, 2, 3, 4\}$. Relabel the points so that*

$$d_{13} + d_{24} \geq d_{12} + d_{34}, \quad d_{13} + d_{24} \geq d_{14} + d_{23}. \quad (8)$$

Define

$$u = \frac{d_{13} + d_{24} - d_{14} - d_{23}}{2}, \quad v = \frac{d_{13} + d_{24} - d_{12} - d_{34}}{2}, \quad (9)$$

$$\begin{aligned} \ell_1 &= \frac{d_{12} + d_{14} - d_{24}}{2}, & \ell_2 &= \frac{d_{12} + d_{23} - d_{13}}{2}, \\ \ell_3 &= \frac{d_{23} + d_{34} - d_{24}}{2}, & \ell_4 &= \frac{d_{14} + d_{34} - d_{13}}{2}. \end{aligned} \quad (10)$$

Then $u, v, \ell_1, \dots, \ell_4 \geq 0$ and the six distances are exactly the ℓ^1 distances obtained from a rectangle of side lengths u, v with pendant length ℓ_i attached to corner i .

Proof. The inequalities (8) give $u, v \geq 0$. Each ℓ_i is a triangle Gromov product; for example $2\ell_1 = d_{12} + d_{14} - d_{24} \geq 0$ by the triangle inequality, and similarly for the others. Direct substitution gives

$$\begin{aligned} d_{12} &= \ell_1 + \ell_2 + u, & d_{34} &= \ell_3 + \ell_4 + u, \\ d_{14} &= \ell_1 + \ell_4 + v, & d_{23} &= \ell_2 + \ell_3 + v, \\ d_{13} &= \ell_1 + \ell_3 + u + v, & d_{24} &= \ell_2 + \ell_4 + u + v. \end{aligned}$$

These are precisely the ℓ^1 distances among the four rectangle corners after adding one private pendant coordinate of length ℓ_i to point i . $\square$

Put

$$x = e^{-u}, \quad y = e^{-v}, \quad r_i = e^{-\ell_i}, \quad D = \text{diag}(r_1, r_2, r_3, r_4),$$

and define the rectangle kernel

$$C = \begin{pmatrix} 1 & x & xy & y \\ x & 1 & y & xy \\ xy & y & 1 & x \\ y & xy & x & 1 \end{pmatrix}. \quad (11)$$

Then the exponential metric kernel is

$$K = DCD + I - D^2. \quad (12)$$

The term $I - D^2$ restores the unit diagonal after pendant attenuation.

For completeness, K is positive definite for a genuine four-point metric. Indeed, lemma 3.1 embeds the metric isometrically into a finite-dimensional ℓ^1 space, and the Laplace kernel $e^{-\|z-z'\|_1}$ is strictly positive definite on distinct points. One direct proof uses its strictly positive Fourier density: for real coefficients c_i ,

$$\sum_{i,j} c_i c_j e^{-\|z_i - z_j\|_1} = C_m \int_{\mathbb{R}^m} \left| \sum_i c_i e^{i\omega \cdot z_i} \right|^2 \prod_{k=1}^m \frac{d\omega_k}{1 + \omega_k^2},$$

where $C_m = \pi^{-m} > 0$. If the integral vanishes, continuity makes the exponential sum vanish at every ω . Choose a vector h for which the real numbers $h \cdot z_i$ are pairwise distinct (the excluded vectors form finitely many hyperplanes). Substituting $\omega = th$ and differentiating at $t = 0$ gives a Vandermonde system, so every $c_i = 0$. This proves strict positive definiteness, including degenerate rectangle representations of distinct metric points.

We now prove the logarithmic sign theorem. The nondegenerate case $0 < x, y < 1$ is treated first; degenerate rectangle limits follow by continuity.

Lemma 3.2 (Adjacent resolvent sign). *Let K be given by (12) with $0 < x, y < 1$ and $0 < r_i \leq 1$. If i, j are adjacent rectangle vertices, then*

$$(K + tI)_{ij}^{-1} < 0 \quad (t \geq 0).$$

Consequently $[\log K]_{ij} > 0$.

Proof. By symmetry it suffices to take $(i, j) = (1, 2)$. Set

$$p = r_3^2, \quad q = r_4^2, \quad A = 1 + t.$$

A cofactor expansion gives

$$(K + tI)_{12}^{-1} = -\frac{r_1 r_2 x Q_{12}(A)}{\det(K + tI)}, \quad (13)$$

where

$$Q_{12}(A) = A^2 - y^2(p + q)A + pq[y^2 - x^2(1 - y^2)]. \quad (14)$$

Since $p, q \leq 1$ and $A \geq 1$,

$$Q'_{12}(A) = 2A - y^2(p + q) \geq 2(1 - y^2) > 0.$$

Thus $Q_{12}(A) \geq Q_{12}(1)$. The latter is bilinear in (p, q) , so its minimum on $[0, 1]^2$ occurs at a corner. The four corner values are

$$1, \quad 1 - y^2, \quad 1 - y^2, \quad (1 - x^2)(1 - y^2),$$

all positive. Since $K + tI \succ 0$, the determinant in (13) is positive. Hence the resolvent entry is negative. Equation (7) gives the logarithm sign. $\square$

The opposite pairs require one additional reduction.

Lemma 3.3 (Complementary-pendant monotonicity). *Let K satisfy (12) with $0 < x, y < 1$ and $0 < r_i \leq 1$, let $L = \log K$, and put $p_k = r_k^2$. Fix an opposite pair i, j . If k is either of the two complementary vertices, then*

$$\frac{\partial L_{ij}}{\partial p_k} < 0.$$

Therefore L_{ij} is minimised, with the target pendants fixed, when both complementary pendant factors equal 1.

Proof. Let $E_k = e_k e_k^T$ and $R_t = (K + tI)^{-1}$. Differentiating (12) with respect to p_k gives

$$\frac{\partial K}{\partial p_k} = \frac{E_k K + K E_k - 2E_k}{2p_k}. \quad (15)$$

For the matrix logarithm, differentiating under the resolvent integral gives the following formula. To justify this step, restrict the parameter to a compact neighbourhood on which the smallest eigenvalue of K stays positive. The derivative integrand is uniformly bounded near zero and has norm $O(t^{-2})$ at infinity. At an endpoint $p_k = 1$, use the resulting one-sided limit.

$$\frac{\partial L}{\partial p_k} = \int_0^\infty R_t \frac{\partial K}{\partial p_k} R_t dt.$$

Using $KR_t = R_t K = I - tR_t$, equation (15) becomes

$$R_t \frac{\partial K}{\partial p_k} R_t = \frac{R_t E_k + E_k R_t - 2(t+1)R_t E_k R_t}{2p_k}.$$

For $i, j \neq k$, the first two terms have zero (i, j) entry, so

$$\frac{\partial L_{ij}}{\partial p_k} = -\frac{1}{p_k} \int_0^\infty (t+1)(R_t)_{ik}(R_t)_{kj} dt. \quad (16)$$

If i, j are opposite and k is complementary, both pairs (i, k) and (k, j) are adjacent. By lemma 3.2, both resolvent factors in (16) are negative for every $t \geq 0$. Their product is positive, giving the claimed strict decrease. $\square$

We can now close the opposite-pair sign exactly.

Lemma 3.4 (Opposite logarithm sign). *For K in (12) with $0 < x, y < 1$ and $0 < r_i \leq 1$, every opposite-pair entry of $\log K$ is nonnegative.*

Proof. By symmetry consider $(i, j) = (1, 3)$. By lemma 3.3, it suffices to set the complementary pendant factors $r_2 = r_4 = 1$. Write

$$a = r_1^2, \quad b = r_3^2, \quad X = x^2, \quad Y = y^2, \quad c = (1 - X)(1 - Y) > 0.$$

A direct cofactor calculation gives

$$(K + tI)_{13}^{-1} = \sqrt{ab}xy \frac{c - t^2}{\Delta(t)}, \quad (17)$$

where $\Delta(t) = \det(K + tI) > 0$. Hence

$$[\log K]_{13} = \sqrt{ab}xy I, \quad I = \int_0^\infty \frac{t^2 - c}{\Delta(t)} dt. \quad (18)$$

The integrand in (18) is bounded near zero and is $O(t^{-2})$ at infinity, so it is absolutely integrable. Let $r = \sqrt{c}$. Splitting the integral at r and applying $t = c/u$ on the upper interval yields

$$I = \int_0^r (c - u^2) \left[\frac{c^2}{u^4 \Delta(c/u)} - \frac{1}{\Delta(u)} \right] du. \quad (19)$$

Thus it is enough to prove

$$F(u) := c^2\Delta(u) - u^4\Delta(c/u) \geq 0 \quad (0 < u \leq r). \quad (20)$$

Set $\alpha = 1 - a$ and $\beta = 1 - b$. Exact expansion gives the factorisation

$$F(u) = (c - u^2) \left[C(u^2 + c) + 2cBu \right], \quad (21)$$

where

$$B = (\alpha + \beta)(X + Y - 2XY) + \alpha\beta XY, \quad (22)$$

$$C = (\alpha + \beta)c(X + Y - XY) + \alpha\beta \left[(X - Y)^2 + XY(3 - 2X - 2Y + XY) \right]. \quad (23)$$

Every factor is nonnegative on $0 < X, Y, a, b \leq 1$. Indeed,

$$X + Y - 2XY = X(1 - Y) + Y(1 - X) \geq 0,$$

and

$$3 - 2X - 2Y + XY = (1 - X)(1 - Y) + (1 - X) + (1 - Y) \geq 0.$$

Therefore $B, C \geq 0$, and on $0 < u \leq \sqrt{c}$ also $c - u^2 \geq 0$. Hence (20) holds. The bracket in (19) is nonnegative, so $I \geq 0$ and (18) gives $[\log K]_{13} \geq 0$. $\square$

For reproducibility, the determinant polynomial underlying (21) is recorded in section A; no numerical approximation is used in the sign argument.

Lemma 3.5 (Metrics on at most three points). *Let (X, d) be a metric space with $2 \leq |X| \leq 3$, and let $K_{ij} = e^{-d_{ij}}$. Then every off-diagonal entry of $\log K$ is strictly positive.*

Proof. For two points take $\ell_1 = \ell_2 = d_{12}/2$. For three points take $\ell_i = (d_{ij} + d_{ik} - d_{jk})/2$ with $\{i, j, k\} = \{1, 2, 3\}$. The triangle inequalities make these lengths nonnegative and $d_{ij} = \ell_i + \ell_j$ for $i \neq j$. This is an isometric embedding into ℓ^1 using private pendant coordinates; hence the preceding Fourier argument makes K positive definite. Put $r_i = e^{-\ell_i}$ and $r = (r_i)$. Then

$$K = \text{diag}(1 - r_i^2) + rr^T.$$

For $t > 0$, set $D_t = \text{diag}(1 + t - r_i^2) \succ 0$. The Sherman–Morrison formula gives, for $i \neq j$,

$$(K + tI)_{ij}^{-1} = -\frac{(r_i/(D_t)_{ii})(r_j/(D_t)_{jj})}{1 + \sum_k r_k^2/(D_t)_{kk}} < 0.$$

The integral representation

$$\log K = \int_0^\infty \left(\frac{1}{1+t} I - (K + tI)^{-1} \right) dt$$

therefore yields $[\log K]_{ij} > 0$. The one-point case is trivial. $\square$

Theorem 3.6 (Four-point theorem). *Let (X, d) be any metric space with $|X| \leq 4$, and let $K(d)$ be defined by (1). Then $K(d)$ is positive definite and*

$$[\log K(d)]_{ij} \geq 0 \quad (i \neq j).$$

Therefore $K(d)$ is strongly infinitely divisible in the ordinary matrix-root sense. Its Perron-normalised stochastic matrix is homogeneously Markov embeddable.

Proof. For $|X| \leq 3$, use lemma 3.5. For four points, use lemma 3.1. In the nondegenerate rectangle case, lemmas 3.2 and 3.4 cover every off-diagonal entry. Degenerate rectangle cases are obtained by replacing the nonnegative side lengths u, v by $u + \varepsilon, v + \varepsilon$ while keeping the pendant lengths fixed. The resulting kernels converge to K as $\varepsilon \downarrow 0$. The limiting K remains positive definite because the original metric points are distinct; continuity of the principal logarithm on that cone therefore preserves the nonnegative signs. The positive-definiteness statement follows from the ℓ^1 representation and the strict positive definiteness of the Laplace kernel on distinct points, as noted above.

Since $B = \log K(d)$ is Metzler, $K(d) = e^B$ and $e^{B/m} \geq 0$ for every $m \geq 1$, proving strong infinite divisibility. Markov embeddability after Perron normalisation follows from lemma 2.3. $\square$

Remark 3.7 (What the theorem does and does not say). *The assertion in theorem 3.6 is about the ordinary matrix logarithm. It supplies information not obtained merely by knowing that the entrywise powers $K^{\text{ot}} = (e^{-td_{ij}})$ are positive semidefinite. No implication between these two divisibility notions is asserted for arbitrary matrices. The latter belongs to Schoenberg–Horn negative-type/Hadamard theory [3, 4]; the former produces ordinary nonnegative matrix roots and a continuous-time Markov generator after Perron normalisation.*

4 A universal five-state deleted-corner obstruction

The four-point result fails sharply at five points. Let

$$B_x = \begin{pmatrix} 1 & x \\ x & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & u & w & uw \\ u & 1 & uw & w \\ w & uw & 1 & u \\ uw & w & u & 1 \end{pmatrix},$$

where the four product states are ordered as $(0, 0), (1, 0), (0, 1), (1, 1)$. Thus C is permutation-similar to $B_u \otimes B_w$. Let e select $(1, 0)$ and define

$$K_{u,v,w} = \begin{pmatrix} C & vCe \\ ve^T C & 1 \end{pmatrix}, \quad 0 < u, v, w < 1. \quad (24)$$

Equivalently,

$$K_{u,v,w} = \begin{pmatrix} 1 & u & w & uw & uv \\ u & 1 & uw & w & v \\ w & uw & 1 & u & uvw \\ uw & w & u & 1 & vw \\ uv & v & uvw & vw & 1 \end{pmatrix}. \quad (25)$$

This is the exponential metric kernel on five retained vertices of a 3×2 product grid: the sixth corner $(2, 1)$ is deleted, while $(2, 0)$ is the fifth retained state. If the consecutive horizontal

lengths are $-\log u$ and $-\log v$ and the vertical length is $-\log w$, then every entry of (25) is $e^{-d_{ij}}$.

Lemma 4.1 (Strict logarithmic obstruction). *Let $a = (0, 0)$ and $b = (1, 1)$ be the opposite retained corners in the original 2×2 square. Then*

$$[\log K_{u,v,w}]_{ab} < 0 \quad \text{for every } 0 < u, v, w < 1.$$

Proof. First, $C \succ 0$ because its eigenvalues are $(1 \pm u)(1 \pm w) > 0$, and the Schur complement of C in (24) is $1 - v^2 e^T C e = 1 - v^2 > 0$. Thus $K_{u,v,w} \succ 0$, justifying the logarithm and the resolvents used below. Set $K_0 = C \oplus [1]$. Since C is permutation-similar to $B_u \otimes B_w$, its logarithm is the same permutation of the Kronecker sum

$$(\log B_u) \otimes I + I \otimes (\log B_w).$$

The entry joining opposite square corners is therefore zero.

For $t > 0$ put

$$A_t = C + tI, \quad x(t) = A_t^{-1} C e, \quad s(t) = 1 + t - v^2 e^T C A_t^{-1} C e.$$

Because $K_{u,v,w} + tI \succ 0$, the scalar Schur complement satisfies $s(t) > 0$. The block inverse formula gives

$$[(K_{u,v,w} + tI)^{-1}]_{OO} = A_t^{-1} + \frac{v^2}{s(t)} x(t) x(t)^T. \quad (26)$$

Subtract the $v = 0$ case in the logarithm integral (6). Since the target entry of $\log K_0$ is zero,

$$[\log K_{u,v,w}]_{ab} = -v^2 \int_0^\infty \frac{x_a(t) x_b(t)}{s(t)} dt. \quad (27)$$

Writing $D(t) = \det(C + tI) > 0$, direct inversion gives

$$x_a(t) = \frac{tu [t^2 + 2(1 - w^2)t + (1 - u^2)(1 - w^2)]}{D(t)} > 0, \quad (28)$$

$$x_b(t) = \frac{tw [t^2 + 2(1 - u^2)t + (1 - u^2)(1 - w^2)]}{D(t)} > 0. \quad (29)$$

Thus the integrand in (27) is strictly positive for $t > 0$. It is $O(t^2)$ near zero and $O(t^{-3})$ at infinity, because $x_a, x_b = O(t)$ at zero, $x_a, x_b = O(t^{-1})$ at infinity, $s(0) = 1 - v^2 > 0$, and $s(t) \sim t$ at infinity. The integral is finite and strictly positive, proving the negative sign. $\square$

The sign in lemma 4.1 alone is not yet enough for general non-embeddability. We now close the alternative-logarithm loophole.

Lemma 4.2 (Simple positive spectrum). *For every $0 < u, v, w < 1$, the matrix $K_{u,v,w}$ is positive definite and has simple positive spectrum.*

Proof. The Schur complement of C in (24) is

$$1 - v^2 e^T C e = 1 - v^2 > 0,$$

while $C \succ 0$, so $K_{u,v,w} \succ 0$.

Diagonalise C in the Walsh basis. Its four eigenvalues are

$$(1+u)(1+w), \quad (1+u)(1-w), \quad (1-u)(1+w), \quad (1-u)(1-w). \quad (30)$$

The coordinate vector e has nonzero projection onto every Walsh vector, and multiplication by C preserves nonzero projections because all four eigenvalues are positive. Hence in the Walsh basis (24) is a real symmetric arrowhead matrix with nonzero coupling from the fifth coordinate to each of the four Walsh basis vectors.

For completeness, an arrowhead matrix A with distinct pole values $d_1 < \dots < d_m$, nonzero couplings z_j , and last diagonal value δ has secular function

$$\phi(\lambda) = \delta - \lambda - \sum_{j=1}^m \frac{z_j^2}{d_j - \lambda}, \quad \phi'(\lambda) = -1 - \sum_{j=1}^m \frac{z_j^2}{(d_j - \lambda)^2} < 0.$$

Its limits at consecutive poles, and at infinity, give exactly one zero in each of the $m+1$ complementary intervals. None of the poles is an eigenvalue: the characteristic polynomial at d_j is $-z_j^2 \prod_{k \neq j} (d_k - d_j) \neq 0$ in the convention $\det(A - \lambda I)$. Hence all the eigenvalues are simple and strictly interlace the poles. No distinctness assumption on δ is needed.

If $u \neq w$, the four numbers in (30) are pairwise distinct. Applying this argument with $m = 4$ proves simplicity.

If $u = w$, the only base degeneracy is the two-dimensional eigenvalue $1 - u^2$. Split that eigenspace into the direction parallel to the coupling vector and its orthogonal complement. The orthogonal direction remains a single eigenvector with eigenvalue $1 - u^2$. The parallel direction enters a reduced arrowhead problem whose three pole values and final diagonal value are, respectively,

$$(1+u)^2, \quad 1 - u^2, \quad (1-u)^2, \quad 1.$$

The three poles are pairwise distinct for $0 < u < 1$, and all reduced couplings are nonzero. The argument just given with $m = 3$ produces four further simple eigenvalues, none equal to any pole and therefore none equal to the uncoupled value $1 - u^2$. Thus the full five-state spectrum is simple. Positive definiteness already proved makes all five eigenvalues positive. $\square$

Theorem 4.3 (Deleted-corner non-embeddability). *Let $0 < u, v, w < 1$ and let $K_{u,v,w}$ be given by (24). Let T be its Perron normalisation. Then T is a strictly positive reversible stochastic matrix with simple positive spectrum, but T is not embeddable in any homogeneous continuous-time Markov semigroup.*

Proof. Reversibility and stochasticity follow from lemma 2.3; simple positive spectrum follows from lemma 4.2. By lemma 2.4, T has a unique real logarithm. Equation (5) and lemma 4.1 show that the (a, b) entry of this unique logarithm is negative. A Markov generator must have nonnegative off-diagonal entries, so no Markov generator can exponentiate to T . $\square$

Corollary 4.4 (Failure of strong infinite divisibility). *For $0 < u, v, w < 1$, the deleted-corner kernel $K_{u,v,w}$ is not strongly infinitely divisible in the ordinary matrix-root sense.*

Proof. Let m be odd and suppose a real matrix A satisfies $A^m = K_{u,v,w}$. Then A commutes with $K_{u,v,w}$. By lemma 4.2, the latter has one-dimensional real eigenspaces, so A is diagonal in the same real eigenbasis. Each eigenvalue of A must be the unique real m th root of the

corresponding positive eigenvalue of $K_{u,v,w}$. Consequently $A = \exp((\log K_{u,v,w})/m)$. For the obstructing pair,

$$A_{ab} = \frac{[\log K_{u,v,w}]_{ab}}{m} + O(m^{-2}).$$

The leading coefficient is strictly negative by lemma 4.1. Thus for every sufficiently large odd m , the only real m th root has a negative entry and cannot be nonnegative. Strong infinite divisibility is impossible. This argument does not use the converse part of the general exponential characterisation. $\square$

5 The exact 4/5 threshold and the divisor-overlap family

Theorem 5.1 (Minimal obstruction cardinality). *Among exponential metric kernels $K(d)_{ij} = e^{-d_{ij}}$, the smallest cardinality for which the Perron-normalised stochastic matrix can fail homogeneous Markov embeddability is five.*

Proof. By theorem 3.6, every exponential metric kernel on at most four points is embeddable after Perron normalisation. By theorem 4.3, every interior member of the five-point deleted-corner family is non-embeddable after Perron normalisation. Hence the minimum is exactly five. $\square$

We now specialise to (2). For one prime axis, if the two exponents are $a \geq b$ then

$$\frac{\min(a, b) + 1}{\sqrt{(a+1)(b+1)}} = \sqrt{\frac{b+1}{a+1}} = \exp\left[-\frac{1}{2} |\log(a+1) - \log(b+1)|\right].$$

Multiplication over prime axes gives (3).

Corollary 5.2 (Infinite arithmetic five-state family). *Let $p \neq q$ be primes, let $k, l \in \mathbb{Z}_{\geq 0}$, and let $c \in \mathbb{N}_{>0}$ be coprime to pq . Define*

$$S_{k,l,c} = c\{p^k q^l, p^{k+1} q^l, p^k q^{l+1}, p^{k+1} q^{l+1}, p^{k+2} q^l\}. \quad (31)$$

Let $Q_S = (q(n, m))_{n, m \in S_{k,l,c}}$ with q as in (2). Then the Perron-normalised Q_S is not homogeneously Markov embeddable.

Proof. The common factor c cancels from every normalised overlap because it is coprime to pq . On the p axis set

$$u = \sqrt{\frac{k+1}{k+2}}, \quad v = \sqrt{\frac{k+2}{k+3}}, \quad (32)$$

and on the q axis set

$$w = \sqrt{\frac{l+1}{l+2}}.$$

The five-state overlap matrix is exactly $K_{u,v,w}$ after a simultaneous permutation of rows and columns. Apply theorem 4.3. $\square$

For $p = 2$, $q = 3$, $k = l = 0$ and $c = 1$, equation (31) gives the original arithmetic restriction

$$S = \{1, 2, 3, 4, 6\},$$

with

$$u = w = 2^{-1/2}, \quad v = \sqrt{2/3}.$$

The present argument proves genuine non-embeddability: the negative principal-logarithm entry is paired with simple positive spectrum, so there is no alternative real logarithm that could serve as a generator.

6 A unique attenuation crossing of the obstructing entry

The five-state obstruction is exact but not purely combinatorial. Independent pendant perturbations can change its sign, so we do not claim that every metric with the same unweighted deleted-corner incidence pattern is obstructed. One controlled perturbation does admit an exact analysis.

Let $K = K_{u,v,w}$ and define

$$K_s = (1 - s)I + sK, \quad 0 \leq s \leq 1. \quad (33)$$

For $0 < s \leq 1$ this is again a strictly positive exponential metric kernel: all off-diagonal distances are increased by the nonnegative constant $-\log s$, which preserves the triangle inequalities. The endpoint $s = 0$ is the identity matrix and is only a limiting kernel; it does not represent finite distances between distinct points. Let a, b be the opposite retained square corners from lemma 4.1, and put

$$f(s) = [\log K_s]_{ab}.$$

Theorem 6.1 (Unique sign crossing under common attenuation). *For every $0 < u, v, w < 1$ there exists a unique $s_{\text{crit}} \in (0, 1)$ such that*

$$f(s) > 0 \quad (0 < s < s_{\text{crit}}), \quad f(s_{\text{crit}}) = 0, \quad f(s) < 0 \quad (s_{\text{crit}} < s \leq 1).$$

Proof. Because K_s is an affine polynomial in K , it commutes with $K - I$. Hence

$$\frac{d}{ds} \log K_s = K_s^{-1}(K - I) = \frac{I - K_s^{-1}}{s}.$$

For $a \neq b$,

$$f'(s) = -\frac{1}{s}(K_s^{-1})_{ab}. \quad (34)$$

Set

$$U = 1 - u^2, \quad V = 1 - v^2, \quad W = 1 - w^2.$$

An exact cofactor calculation yields

$$(K_s^{-1})_{ab} = -\frac{suw H(s)}{\det K_s}, \quad (35)$$

where

$$\begin{aligned} H(s) = & 1 + (V - 3)s + (UVW - 2UW - 2V + 3)s^2 \\ & + (-2UVW + 2UW + V - 1)s^3. \end{aligned} \quad (36)$$

Since $K_s \succ 0$ for $s \in (0, 1]$, equations (34)–(35) show that $\text{sgn } f'(s) = \text{sgn } H(s)$.

Write H in the Bernstein basis on $[0, 1]$:

$$H(s) = b_0(1-s)^3 + 3b_1s(1-s)^2 + 3b_2s^2(1-s) + b_3s^3,$$

with

$$b_0 = 1, \quad b_1 = \frac{V}{3}, \quad b_2 = \frac{UW(V-2)}{3}, \quad b_3 = -UVW. \quad (37)$$

For $U, V, W \in (0, 1)$ the signs are $+, +, -, -$, so the Bernstein coefficient sequence has exactly one sign variation. One may establish the required root count directly, without a separate variation-diminishing theorem. Put $z = s/(1-s) > 0$. Then

$$P(z) := (1+z)^3 H\left(\frac{z}{1+z}\right) = 1 + Vz + UW(V-2)z^2 - UVWz^3.$$

The function

$$\frac{P(z)}{z^2} = z^{-2} + Vz^{-1} + UW(V-2) - UVWz$$

is strictly decreasing, since its derivative is $-2z^{-3} - Vz^{-2} - UVW < 0$, and its limits at zero and infinity are $+\infty$ and $-\infty$. It has exactly one positive zero, so H has exactly one zero in $(0, 1)$. Consistently,

$$H(0) = 1 > 0, \quad H(1) = -UVW < 0,$$

so H has exactly one zero s_* . Thus f first increases and then decreases. Since $f(0) = 0$, it is positive on $(0, s_*]$. By lemma 4.1, $f(1) < 0$. Strict decrease after s_* forces a unique zero $s_{\text{crit}} \in (s_*, 1)$ with the stated sign pattern. $\square$

Remark 6.2 (What the crossing proves). *For sufficiently small $s > 0$, all off-diagonal entries of $\log K_s$ are positive because*

$$\log(I + s(K - I)) = s(K - I) + O(s^2)$$

and K is strictly positive off diagonal. This is a local expansion, valid for $s\|K - I\|_2 < 1$; it is not an assertion that the Mercator series converges at $s = 1$. Moreover, for $s > 0$ the eigenvalues $1 - s + s\lambda_j(K)$ remain positive and pairwise distinct. Thus the Perron-normalised kernel is embeddable near $s = 0$. For $s > s_{\text{crit}}$, the negative entry in theorem 6.1, together with simple positive spectrum of K_s , certifies non-embeddability. We do not claim here that no other logarithm entry can become obstructive before s_{crit} ; establishing a complete embeddability phase diagram would require an additional argument.

7 Scope, failure modes excluded, and current frontier

The proof architecture above is deliberately narrower than a general study of finite metric kernels or Markov embedding. The following boundaries are part of the result.

- (i) **Principal-log failure is not generic non-embeddability.** A negative off-diagonal entry of the principal logarithm is insufficient for a general Markov matrix [9]. The five-state theorem therefore includes the simple-positive-spectrum proof that makes the real logarithm unique.

- (ii) **Ordinary matrix divisibility is not Hadamard kernel divisibility.** The logarithm in this paper is the matrix function $\log K$, not the entrywise matrix $(-\log K_{ij})$. The roots are ordinary matrix roots, not entrywise powers. Classical Schoenberg–Horn results remain relevant background, but they answer a different divisibility question [3, 4].
- (iii) **The five-state theorem uses exact product geometry.** The deleted-corner obstruction is not asserted to survive arbitrary independent pendant attenuation of the retained states. Such perturbations can change the obstructing logarithm sign. The universal claim is exactly theorem 4.3, with the parameterisation (24).
- (iv) **The four-state proof is not an inverse- M -matrix argument.** Some four-point exponential metric kernels have inverse matrices with positive opposite off-diagonal entries. The proof therefore works directly with the logarithm through resolvents, pendant monotonicity, and the reciprocal factorisation (21).
- (v) **No complete five-state classification is claimed.** Theorem 5.1 determines the first cardinality at which an obstruction can occur, not which five-point metrics are obstructed. There are five-state kernels outside the deleted-corner family whose status requires separate analysis.
- (vi) **Current cube evidence is not promoted to theorem.** Five retained vertices of a $2 \times 2 \times 2$ product cube fall into three symmetry orbits. The deleted-corner theorem covers one orbit. The other two have strong computational evidence of obstruction, but no analytic proof is included here.
- (vii) **No priority claim is made from absence of a search hit.** The literature already contains complete general 4×4 embedding criteria [10], reversible embedding theory [6, 12], strong infinite-divisibility theory [7, 13], and recent stochastic-root refinements [11]. The contribution stated here is the explicit metric-kernel theorem chain proved in the paper; questions of priority for equivalent formulations should be assessed against that literature.

The threshold established by theorem 5.1 leaves the internal geometry of order five unresolved. Three questions are left open by the present paper: characterising all five-point exponential metric kernels with Metzler logarithm; proving or refuting universal obstruction for the two unresolved five-of-eight cube orbits; and finding an intrinsic metric invariant that predicts the first negative logarithm entry without computing the full matrix logarithm.

8 Conclusion

Exponential metric kernels sit at the intersection of finite metric geometry, nonnegative matrix roots, and the Markov embedding problem. At four points this intersection is rigid: every metric admits a rectangle-plus-pendants representation whose exponential kernel has a Metzler matrix logarithm. At five points the rigidity fails, and it fails in a fully explicit product-metric family. The simple-spectrum step is essential: it upgrades a principal-logarithm sign observation into a genuine obstruction to every homogeneous Markov embedding.

The resulting threshold is exact. Four states always embed after Perron normalisation; five states can fail. The divisor-overlap construction supplies an infinite arithmetic family of such failures, while common attenuation of the deleted-corner family exhibits a single exact sign

transition of the obstructing logarithm entry. Beyond these statements, the five-state classification remains open, and no stronger classification is inferred from the present computations.

A Algebraic certificate for the reciprocal factorisation

In the reduced opposite-pair family used in lemma 3.4, write $a = r_1^2$, $b = r_3^2$, $X = x^2$, $Y = y^2$, and $c = (1 - X)(1 - Y)$. With $\Delta(t) = \det(K + tI)$, direct determinant expansion gives

$$\begin{aligned}\Delta(t) = & t^4 + 4t^3 \\ & + t^2(-abXY - aX - aY - bX - bY - XY + 6) \\ & + t(2abXY + 2aXY - 2aX - 2aY + 2bXY - 2bX - 2bY - 2XY + 4) \\ & + abX^2Y^2 - 2abX^2Y + abX^2 - 2abXY^2 + abXY + abY^2 \\ & + 2aXY - aX - aY + 2bXY - bX - bY - XY + 1.\end{aligned}$$

Set $\alpha = 1 - a$, $\beta = 1 - b$ and define B, C by (22)–(23). Clearing the reciprocal argument gives the exact polynomial identity

$$c^2\Delta(t) - t^4\Delta(c/t) = (c - t^2)[C(t^2 + c) + 2cBt].$$

No limiting or floating-point argument is used in lemma 3.4; this identity can be checked by ordinary polynomial expansion.

B Status of the earlier arithmetic example

For the arithmetic restriction $S = \{1, 2, 3, 4, 6\}$, an earlier derivation correctly found a negative off-diagonal entry of the Perron-normalised principal logarithm but did not, by that fact alone, exclude a different real logarithm from being a Markov generator. The present paper repairs that gap in two ways. First, S is a special case of the universal deleted-corner family in corollary 5.2. Second, lemma 4.2 proves simple positive spectrum for the whole family, so lemma 2.4 makes the principal logarithm the unique real logarithm. The non-embeddability conclusion for S is therefore now branch-independent.

C Proof status and reproducibility

The results in this article are written mathematical proofs. The word “proved” in the main text refers to those arguments, not to a claim that the entire theorem chain has been accepted by a proof assistant. In particular, the arbitrary-metric-to-kernel construction, the matrix-logarithm analytic steps, and the concrete spectral and generator instantiations have not been assembled into an end-to-end Lean theorem in the materials available for this revision.

The recoverable 17 September 2026 supplement contains one module, `MarkovEmbeddingAudit.lean`, together with historical build logs and a pinned Lean/mathlib configuration. It checks scalar algebra and inequality components: rectangle reconstruction, adjacent-sign polynomials, the reciprocal determinant identity, attenuation coefficients, three-point reconstruction, and deleted-corner numerator positivity. These checks do not by themselves bind every polynomial to its displayed matrix, prove the required integral hypotheses, or deliver the final

Markov-embedding statements. The historical logs report successful compilation; this revision did not rerun Lean.

A separate status review dated 18 September 2026 reports a later build of twelve research modules and a declaration audit covering 106 declarations. The corresponding full source-and-receipt bundle was not recovered for this revision, so those counts are source-reported background and are not transferred to the recoverable one-module supplement. That review also explicitly withholds full formalisation of the headline results.

For this revision, an independent symbolic script recalculates nine exact identities, including cofactors from the displayed four- and five-state matrices, the reduced determinant, reciprocal factorisation, and attenuation polynomial transformations. It also executes finite floating-point regressions on 1,000 four-point shortest-path metrics and 1,025 deleted-corner parameter triples, including 25 equal-axis cases. These tests supplement the proofs; they do not prove universal assertions. The script and its machine-readable results are supplied with the candidate.

For orientation, in the arithmetic ordering $(1, 2, 3, 6, 4)$, numerical eigendecomposition gives $[\log K]_{1,6} \approx -0.0359491773892$ and the designated attenuation crossing $s_{\text{crit}} \approx 0.979464557852$. These are reproducible numerical illustrations, not interval-certified values. The proof of existence and uniqueness of the crossing is theorem 6.1; it is not a whole-generator phase boundary.

The final deposition still requires the author’s choice of licence and confirmation of version relationships. No DOI is assigned to this corrected candidate in the manuscript.

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